

Daniel Mercado-Bettín, PhD

Hydrologist · Hybrid Process-Based / AI Modelling · Operational Forecasting Systems

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ORCID: orcid.org/0000-0003-4572-3029 · Web of Science ResearcherID: HPF-6131-2023 · Portfolio: danielmerbet.github.io

PROFILE

Hydrologist with 8+ years' postdoctoral experience building hybrid process-based and machine-learning models across hydrology, hydrogeology and limnology, and deploying them operationally for real institutional users. My work spans seasonal streamflow and reservoir forecasting, groundwater modelling, flood hydraulics, and global-scale climate-impact simulation on HPC infrastructure. I have led international research programmes, secured competitive funding, and supervised postgraduate researchers, with a demonstrated ability to translate rigorous, skill-verified science into tools that non-technical decision-makers actually use.

CORE COMPETENCIES

Hybrid process-based & AI hydrological modelling · Groundwater modelling (MODFLOW 6, FloPy) · Hydraulic & flood modelling (ANUGA) · Operational seasonal & sub-seasonal forecasting · Remote sensing & GIS (MODIS, Sentinel, LiDAR) · Large-scale HPC simulation · Research funding acquisition & project leadership · Postgraduate supervision & researcher training · Python & R programming · Peer-reviewed scientific publication

PROFESSIONAL EXPERIENCE

Scientist & Modeller (Postdoctoral Researcher) *Jun 2018 – present*

Spanish National Research Council (CSIC-CEAB), Blanes & Catalan Institute for Water Research (ICRA), Girona

- **Operational seasonal forecasting:** designed, developed and operate NEREIDA, a live seasonal forecasting tool for the Sau–Susqueda–Pastoral reservoir cascade, built for the Catalan Water Agency and used daily in real operational decisions.
- **Groundwater modelling:** independently developed a regional MODFLOW 6 groundwater model of Catalonia (via FloPy), coupled with seasonal water-table forecasts fit to a public piezometric network of ~2,200 wells.
- **Hydraulic & flood modelling:** built a 2D hydraulic flood model (ANUGA) for the Ter & Onyar rivers through Girona, calibrated against real storm events with formal skill metrics.
- **Global-scale hydrological modelling:** led the ISIMIP global lake sector, running hybrid process-based/ML climate-impact ensembles across 41,000 lakes on HPC infrastructure (>20 million core-hours).
- **Regional forecasting toolkit development:** co-developed WATExR, an open-source seasonal forecasting workflow, applied across 7 European catchments with national water authorities.
- **Funding acquisition & research leadership:** coordinated a €4M Marie Skłodowska-Curie research network (15 researchers, 11 partner institutions), owning delivery and direct reporting to the European Commission; secured €173,000 in competitive funding across 7 international projects.
- **Teaching & outreach:** designed and led hands-on hybrid-modelling workshops (incl. GLEON 2026) and public-engagement activities translating operational forecasting for general audiences; supervised 2 PhD and 3 MSc students.
- Published 22 Web of Science-registered peer-reviewed articles (incl. *Science*, 2× *Nature Communications*, *Ecology Letters*, *Water Research*); presented at 10–20 international conferences.

Predoctoral Researcher *Jan 2014 – Mar 2018*

University of Antioquia, Medellín, Colombia

- Built hydrological impact models spanning local watersheds to large basins (incl. the Amazon), studying the statistical and physical regulation of extreme river flows by forest cover, and integrating global climate and remote sensing datasets.
- PhD awarded Cum Laude with full national scholarship; DAAD-funded research stay at University of Kiel, Germany.

SELECTED TOOLS & DELIVERABLES

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- **NEREIDA — Seasonal Forecast Viewer:** live operational forecasting tool for Iberian reservoirs — nereidaca.github.io/hydro_tool
 - **Catalunya Groundwater Model:** MODFLOW 6/FloPy regional model with seasonal well forecasts — danielmerbet.github.io/gw_catalunya
 - **Girona Flood Model:** ANUGA hydraulic flood model for the Ter & Onyar rivers — danielmerbet.github.io/Floods_Girona-Spain
 - **WATExR Toolkit:** open-source seasonal forecasting workflow co-developed with 7 European water authorities — github.com/icra/WATExR
 - **ISIMIP Global Lake Simulations:** open database of climate-impact ensembles across 41,000 lakes — isimip.org/impactmodels/details/205

EDUCATION

PhD, Environmental Engineering 2014 – 2018

University of Antioquia, Colombia — awarded Cum Laude

BSc, Environmental Engineering 2008 – 2013

University of Antioquia, Colombia

TECHNICAL SKILLS

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- **Programming:** Python, R, MATLAB, Bash, Linux, Git/GitHub, HPC clusters.
 - **Machine learning:** Random Forest, XGBoost, LSTM, neural networks.
 - **Hydrological, hydraulic & groundwater modelling:** GOTM, GLM-AED, mHM, SWAT, GR4J, ANUGA, MODFLOW 6/FloPy.
 - **Weather & climate data:** ECMWF, ERA5, CFSR, SEAS5, CMIP5/6, CORDEX; tools CDO, NCO, GDAL.
 - **Remote sensing & GIS:** MODIS, Landsat, Sentinel, LiDAR; QGIS, GRASS, ArcGIS.

LANGUAGES

Spanish (native) · English (C1, professional).

REFERENCES

Rafael Marcé

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Full publication list: orcid.org/0000-0003-4572-3029 · Web of Science ResearcherID: HPF-6131-2023 · Portfolio & code:

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